UX AUDIT · /ANDROID-UX-AUDIT

Web Listener App

Airwave Agents — `apps/web`

Scope	App · RadioPlayer · ChatBox · Header · ProfileModal · StationDirectoryModal · RjAnnouncementBanner — against `tailwind.config.js` + `index.css` + `index.html`
Findings	22 total — 1 P1 · 5 P2 · 11 P3 · 5 P4
Date	2026-10-15

> **Inferred intent** — *confirm before acting on the P1.* A listener opens the page, usually on a phone, wants to press play and hear the live station, see what's on air, and chat with the AI host. Worst cases: audio fails/stalls silently, the browser blocks autoplay, or the UI claims "live" while the stream is silent.

Summary

SEVERITY	COUNT	THEME
P1	1	The player reports "in sync, playing in real time" while actually silent
P2	5	Hover-trap menu · unnamed transport controls · raw `alert()` errors · country-select desync · listener can edit the raw stream URL
P3	11	Loading/empty conflation · modals without Esc/focus-trap · sub-12px text · sub-44px targets · unused tokens · Inter never loaded · no offline state
P4	5	No char counter · dead `darkMode` flag · Cancel enabled during save · clipboard no-catch · phantom "cooldown"

Platform note: hover *exists* on web, but the app is used heavily on mobile browsers — so ≥44px touch targets, thumb reach, and "nothing critical behind hover on touch" still apply. Back = browser back; modals must handle Esc / overlay-click / focus trapping.

The theme that matters most: false success

A healthy-looking player over a silent stream

The one P1 is the product-wide false-success bug in its most listener-facing form: a positive, reassuring state shown while the real system produced no sound. The fix pattern is the same everywhere — **drive status from the real signal** (playback `onPlaying`, not a timer) and **never render a positive state unless the operation actually succeeded.**

WHERE	SHOWS	REALITY
RadioPlayer `:56`	green "Live Broadcast In Sync · playing in real time"	autoplay was blocked — the user hears nothing and was never told to tap play

P1 — Critical

loses the listener / hides a dead stream

P1-1 **RadioPlayer · UI reports "Live Broadcast In Sync — playing in real time" while silent**

WHERE `RadioPlayer.tsx:56–103` (sync timer) → `407–468` (footer) + `118–145` (autoplay)

PROBLEM The 20-second sync countdown runs on a bare `setInterval` keyed only on `radioStatus.isLive` — it is **not** tied to real playback. When the browser blocks autoplay (the single most common real failure — the `.catch` at `:139` only logs and sets `isBuffering=false`), `syncSeconds` still ticks to 0. The badge then flips to "LIVE ON AIR" and the footer renders the green "…playing in real time · 0s BUFFER LAG • LIVE" while the user hears nothing. The same false positive occurs if `onError` fires (`:239`) after the timer reached 0. This is the audio equivalent of showing "Saved" on a failed write.

FIX Drive the "in sync" state from the actual `onPlaying` / `isPlaying` signal, not the timer. On autoplay reject, set a distinct `needsGesture` state and render **"Tap ▶ to start listening"** instead of the green panel. On `onError` keep the red alert and never show "in sync". Never render "playing in real time" unless `isPlaying && !isBuffering` .

Nielsen N1 (visibility of true status) · Autosave pattern ("never show Saved for a failed write") · Error States

P2 — Major

works on desktop, breaks on a phone + error honesty

P2-1 **Header · Profile dropdown dismiss depends on `onMouseLeave` (Hover Trap)**

WHERE `Header.tsx:185–189`

PROBLEM The menu only auto-closes via `onMouseLeave` , which never fires on touch. On a phone it can be closed only by re-tapping the avatar; tapping elsewhere leaves it stuck open over the content.

FIX Remove the hover dismiss. Add a full-screen click-away backdrop (or outside-click listener) and close on item select / Esc.

Hover Trap (the #1 mobile-web pitfall) · Jakob's Law

P2-2 **RadioPlayer · Core transport controls have no accessible names**

WHERE `RadioPlayer.tsx:318–335` (play/pause), `339–348` (mute), `349–360` (volume), `364–370` (settings)

PROBLEM All are icon-only with a `title` but no `aria-label` . Screen-reader users get "button" with no action; the volume `<input type=range>` is unlabeled. The play button is the screen's primary action.

FIX Add `aria-label` on each (toggle "Play"/"Pause" with state); give the range `aria-label="Volume"` + `aria-valuetext` .

WCAG 4.1.2 (Name/Role/Value) · accessibility ref

P2-3 **App · Sign-in failure uses a raw `alert()` , and cancelled popup is treated as an error**

WHERE `App.tsx:329–336`

PROBLEM `alert("Sign-in failed: " + err.message)` is an unstyled, blocking, raw-message dialog. Closing the Google popup (benign cancellation) hits the same path and scolds the user with a "failed" alert.

FIX Inline styled banner/toast mapped to cause (popup blocked, network, cancelled). Treat `auth/popup-closed-by-user` / `cancelled-popup-request` as a no-op.

CLAUDE.md "Handle every state" (cancellation is benign) · Error States

P2-4 **ProfileModal · Country default `"LK"` desyncs the select from the preview and stores mixed formats**

WHERE `ProfileModal.tsx:23, 126–146`

PROBLEM `country` initializes to code `"LK"` , but `<option value={c.name}>` uses names. No option matches, so the select shows the *first* country while the preview says Sri Lanka. Untouched, `"LK"` is saved; picking one saves the full name — two formats in Firestore for one field.

FIX Default to a value that exists in the options (prefer canonical `code`) and store one form consistently across app / chat / header.

Nielsen N6 (recognition) · Form Field States

P2-5 **RadioPlayer · Raw Icecast stream-URL editing exposed to listeners**

WHERE `RadioPlayer.tsx:364–370, 471–509`

PROBLEM The settings gear lets any listener edit/replace the station's stream URL — an internal transmitter detail a listener should never touch, and one that can break their own playback with no recovery. A data-model field surfaced as UI.

FIX Remove the URL editor from the listener surface (the stream comes from the selected station's Firestore doc). Keep it only behind a dev / query-param flag.

Intent clarity §1.3 ("nothing exists only because the data model has that field") · Simplicity First

P3 — Medium / polish

P3-1 **Loading state conflated with empty state**

WHERE `ChatBox.tsx:110–119` · `StationDirectoryModal.tsx:48–55` · `App.tsx:540–556`

FIX During the first Firestore snapshot the arrays are empty, so "No messages / stations / broadcaster yet" flashes before data arrives. Track a `loaded` flag per listener; show a skeleton until the first snapshot resolves. *Loading States System · Empty States*

P3-2 **Modals have no Esc / backdrop close and no focus trap**

WHERE `ProfileModal.tsx:79–81` · `StationDirectoryModal.tsx:22–24`

FIX Close on Esc + overlay click; trap focus; restore focus to the trigger on close; add `role="dialog"` + `aria-modal` . *Nielsen N3 · WCAG 2.4.3*

P3-3 **Pervasive sub-12px text harms readability**

WHERE `text-[10px]/[11px]` across `App.tsx:505/611/642` · `Header.tsx:92/177` · `RadioPlayer.tsx:275/424` · `ChatBox.tsx:94/153`

FIX Floor body/meta text at ~12–13px; reserve 10px for non-essential decoration. *Aesthetic-Usability vs legibility*

P3-4 **Multiple touch targets below 44px**

WHERE mute `RadioPlayer.tsx:339` · dismiss X `RjAnnouncementBanner.tsx:46` / `StationDirectoryModal.tsx:38` · Copy-Link `Header.tsx:94` · chat voice `ChatBox.tsx:215`

FIX Pad interactive icons to a ≥44×44 hit area. *Fitts's Law · WCAG 2.5.8*

P3-5 **Defined `brand` tokens unused; many competing accents**

WHERE `tailwind.config.js:9–33` vs raw `blue/indigo/purple/emerald/amber/red` in components

FIX Route primary actions through `brand-*` ; one accent per surface; strict semantic roles (red = on-air only, emerald = connected only). *Von Restorff · design-system consistency*

P3-6 **Inter is configured but never loaded**

WHERE `tailwind.config.js:35–44` · no font `<link>` in `index.html`

FIX `fontFamily.sans` leads with "Inter" but nothing loads it, so the app silently falls back to system fonts. Load Inter (Google Fonts link or `@fontsource/inter`) or drop it from the stack. *Visual Character — type axis*

P3-7 **No offline indicator**

WHERE `App.tsx` (no `navigator.onLine`) · `RadioPlayer.tsx:239–243`

FIX Listen to `online` / `offline` ; show "You're offline — reconnecting" and distinguish it from a station-side stream error. *State Completeness (offline) · Error States*

P3-8 **Header logo fallback never renders on image error**

WHERE `Header.tsx:62–72`

FIX On `onError` the `<img>` is hidden, but the fallback `only:flex` never matches (img + fallback + ping coexist), leaving blank space. Toggle the fallback with state instead. *broken-media state*

P3-9 **Sign-out and profile-read failures are silent**

WHERE `App.tsx:338–345` (sign-out) · `134–138` (profile read)

FIX Failed sign-out logs to console only (user still appears signed in); a failed profile read silently appears signed-out. Toast on sign-out failure; retry + minimal authed profile on profile-read error. *Nielsen N1/N9*

P3-10 **Static marketing cards crowd live content on mobile**

WHERE `App.tsx:602–657`

FIX Three static explainer cards push real interaction below a wall of promo copy on mobile. Collapse into one "About this station" disclosure or move below the fold. *Information Hierarchy · de-AI landing*

P3-11 **RJ voice banner can claim "Playing Voice" when autoplay is blocked**

WHERE `RadioPlayer.tsx:159–162` · `RjAnnouncementBanner.tsx:36–45`

FIX `rjAudioRef.play()` may reject, but the banner already shows "Playing Voice." Flip the label on `onplaying`; show "Tap to hear" on rejection. *Nielsen N1*

P4 — Minor

P4-1 **ChatBox · `maxLength={280}` with no character counter**

WHERE `ChatBox.tsx:306` — input silently stops at 280. FIX show a counter near the limit. *Form Field States*

P4-2 **Config · `darkMode:"class"` set but no dark theme or toggle exists**

WHERE `tailwind.config.js:7` — dead config; system-dark users get forced light. FIX implement a dark theme or remove the flag.

P4-3 **ProfileModal · Cancel stays enabled during save**

WHERE `ProfileModal.tsx:150–158` — allows dismissing mid-write. FIX disable Cancel while `saving`.

P4-4 **App / Header · Clipboard writes have no catch/fallback**

WHERE `App.tsx:520` · `Header.tsx:51` — `clipboard.writeText` throws on insecure context/older browsers with no feedback. FIX wrap in try/catch with a "Copy failed" fallback.

P4-5 **ChatBox · Misleading "cooldown" label with no implementation**

WHERE `ChatBox.tsx:261` — intended as a "20-Min Cooldown Section" (`lastChatAt` tracked in `App.tsx:369–372`) but no cooldown UI exists.
FIX implement a visible countdown, or drop the dead affordance and map the rate-limit error.

Coverage checklist

AREA	VERDICT
Intent clarity	**Pass** — listener goal clear; player + chat dominate. Exception: the exposed stream-URL editor (P2-5).
Hierarchy	**Partial** — good primary focus, but static marketing cards crowd live content on mobile (P3-10).
States (loading/empty/partial/error/success/offline)	**Fail** — biggest gaps: the P1 false-sync, loading/empty conflation (P3-1), no offline path (P3-7).
Forms	**Partial** — good validation/labels; holes are the country default (P2-4) and no modal focus/Esc (P3-2).
Feedback & affordance	**Partial** — play/mute a11y (P2-2), hover-trap menu (P2-1), raw `alert()` (P2-3).
Design-system consistency	**Fail** — defined `brand` tokens unused, Inter never loaded, many competing accents (P3-5, P3-6).
Visual character	**Pass** — recognizable radio product; tighten to one accent and actually load Inter to clear generic-AI default.

Suggested fix order

1. **P1-1 first** — tie the player's "in sync / live" state to real playback (`isPlaying`) and add the autoplay-blocked "Tap ▶ to start listening" prompt. This is the one that makes a silent player look healthy.
2. **The mobile-web a11y + hover sweep** — P2-1 (hover-trap menu), P2-2 (transport aria-labels), P3-2 (modal Esc/focus-trap), P3-4 (≥44px targets), P3-3 (sub-12px text). Mostly mechanical.
3. **Error honesty** — P2-3 (`alert()` → mapped banner, cancellation benign), P3-7 (offline banner), P3-9 (silent sign-out/profile failures), P3-1 (skeletons instead of flashing empty).
4. **Design-system + character** — P3-5 (one accent via `brand-*`), P3-6 (load Inter), P2-5 (remove the listener stream-URL editor), P2-4 (fix the country default). One pass, lifts the whole app.